CausalCD: A Causal Graph Contrastive Learning Framework for Self-Supervised SAR Image Change Detection

Haolin Li, Bin Zou, *Senior Member, IEEE*, Lamei Zhang, *Senior Member, IEEE*, and Jiang Qin

***Abstract*— In recent years, self-supervised synthetic aperture radar (SAR) image change detection methods have achieved remarkable results, particularly in reducing dependence on expensive supervised signals. However, some critical issues remain open: how to design the optimal unsupervised feature representation model, and how to utilize noisy pseudo-labels for self-training to improve the change detection model? In this article, efforts are made to find a principled and fundamental solution to the above issues from the new perspective of causal reasoning, proposing a self-supervised SAR change detection framework named CausalCD. Specifically, we first construct a structural causal model (SCM) to formalize the self-supervised SAR image change detection process, and carry out a principled analysis to assess the influence of the unsupervised feature representation module and the pseudo-label-driven self-training module on the performance of change detection model. On this basis, we present a feature representation module that employs graph contrastive learning (GCL) and leverages the causal invariant mechanism to extract the optimal augmented representation, thus improving the model's generalizability and soundness. In addition, we employ a causal intervention strategy to mitigate the negative impact of confusing bias from noisy pseudo-labels by blocking the backdoor path. CausalCD's strengths lie in reducing the model's dependence on labeled samples through causal GCL and obtaining optimal feature representation. Concurrently, it helps disentangle the noisy pseudo-label's impact, thereby improving the performance of the self-supervised SAR change detection model. Finally, comprehensive experiments on three bitemporal SAR scenes demonstrate that CausalCD significantly outperforms several mainstream change detection models, confirming the effectiveness of CausalCD.**

***Index Terms*— Causal inference, change detection, graph contrastive learning (GCL), self-supervised learning (SSL), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has become an indispensable tool in Earth observation due to its all-time and all-weather observation capabilities. SAR image change detection, as a crucial aspect of SAR data interpretation, aims to analyze SAR images of the same area at different times to identify areas where changes have occurred on the surface. This technique plays an essential role in various fields, including disaster early warning and assessment, environmental monitoring, and urban change analysis [1], [2], [3].

The practical significance of SAR image change detection tasks has garnered extensive research interest from the remote sensing community and academia. Supervised-based algorithms that combine learning models with labeled data have yielded fruitful results in change detection. Compared with the traditional change detection approaches, their advantage lies in that they can acquire feature presentation by learning the training data without manually designing the feature extractor, and data-driven change detection frameworks can be constructed [4], [5], [6]. Most learning-based algorithms leverage convolutional neural network (CNN) as the basic framework, which follows the strategies based on image patches [7], [8], [9]. Compared with only considering the feature information of a single pixel, embedding the pixel's neighborhood information into the feature representations can prevent the false detection caused by speckle noise to a certain extent. However, such methods will result in poor edge preservation ability of change detection results. In [10], a framework for SAR image change detection is introduced, selecting spatially homogeneous neighborhoods adaptively and determining the multitemporal windows based on a time-adaptive strategy, thereby improving edge preservation. In this article, we explore improving the edge preservation ability of change detection results from another perspective, i.e., by adopting flexible graph structures instead of regular data structures. The graph feature representation models developed on this basis can generalize the neural network to irregularly structured inputs, map complex graph structures into compact vector representations, and preserve important structural and semantic information, which have become valuable tools to address this challenge [11], [12].

The training of the aforementioned change detection deep networks usually heavily relies on a large number of supervised samples, such as labels for supervised learning or rewards for reinforcement learning. However, due to the complex imaging mode of SAR satellites, there will be inevitable coherent speckle noise and clutter, making it expensive or impractical to manually label the change areas of the SAR images, which has significantly restricted the broad applications of such studies. On the other hand, massive unsupervised

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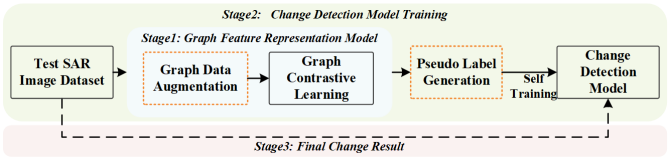


Fig. 1. Pipeline for self-supervised SAR image change detection based on GCL, which is the foundation for our work.

samples are available. Consequently, current research often focuses on designing completely unsupervised SAR image change detection algorithms, where the training samples for changed or unchanged areas are not labeled in any form. For the past several years, self-supervised learning (SSL) strategy has been broadly adopted in unsupervised algorithms, which can utilize rich real-world data while not requiring expensive human-annotated information [13], [14], [15]. Among various SSL techniques, graph contrastive learning (GCL) has risen to prominence as a leading unsupervised method for learning nodes or whole graph representations [16]. Further, to employ the graph feature representation model in the downstream change detection tasks, a self-training strategy of pseudo-label generation is usually employed. The core idea of this strategy is that heuristic-based pseudo-labels can fine-tune the unknown downstream tasks without labels or external rewards, which can reduce the cost of model learning.

Although many SSL change detection models have demonstrated competitive performances, they still have some key issues unresolved due to limited development time.

Q1: The change detection models tend to focus only on the statistical relationships between data and labels, rather than on the causal relationships, which can reduce the models' reliability.

Q2: The performance gain of the pseudo-label-driven self-supervised change detection models mainly comes from correct pseudo-labels, while incorrect pseudo-labels will negatively confuse the model.

Q3: The generation of pseudo-labels relies heavily on heuristic designs and tends to ignore causality, limiting potential enhancements in the performance of change detection models.

To tackle the above issues, we carry out an in-depth analysis of the self-supervised SAR change detection pipeline both theoretically and experimentally. The pipeline for self-supervised SAR image change detection based on GCL is shown in Fig. 1. It consists of three main stages.

1) *Stage 1:* Given multiple unlabeled bitemporal SAR image datasets, a feature representation model is trained by the GCL framework.
2) *Stage 2:* For each test dataset, the corresponding features are obtained by the graph feature representation model obtained in stage 1, and then the pseudo-labels generated by the traditional heuristic method are used to train the change detection model.
3) *Stage 3:* The final change detection results of the test datasets are obtained by the change detection model obtained in stage 2.

Then, we draw two conclusions: 1) employing the traditional graph data augmentation strategy within the graph feature representation model may cause the model to learn irrelevant features, resulting in significant performance degradation and 2) the pseudo-labels generated based on the heuristic design inevitably contain noisy samples, which lead to degraded performance of the change detection model to some extent.

We re-investigate the above problems through the lens of causal inference, and introduce a self-supervised SAR image change detection framework named CausalCD. Specifically, in stage 1, we construct a structural causal model (SCM) to explain the causal invariant mechanism in the GCL framework. Based on this, a new graph data augmentation strategy is designed, which can infer the attributes of the optimal augmented representation to improve the feature representation model's generalizability and soundness. Furthermore, in stage 2, considering the difficulties in manually identifying confounders and their specific impacts within self-supervised training models, we construct an SCM to formalize the pseudo-label-driven self-training process. Then, backdoor adjustment is utilized to disentangle the interference caused by the noisy pseudo-labels to the change detection model. This allows the model to capture the causal relationship between data and labels, thereby improving its robustness and reliability and ultimately achieving superior performance across various complex scenarios.

In summary, this article has the following contributions.

1) The construction of SCM formalizes the process of self-supervised SAR image change detection, which can be used to perform principled analysis of the impact of the feature representation module and the pseudo-label-driven self-training module on the performance of the change detection model. To our knowledge, our method is the first to introduce the improvement based on the causal theory into SAR image change detection.
2) The feature representation process based on GCL is analyzed from a causal inference perspective, leading to the design of a novel graph data augmentation strategy grounded in the causal invariant mechanism. This strategy incorporates constraints based on the graph structure diversity and semantic diversity to guarantee the generation of hard positive and negative instances of the original graph. Such an approach aims to bolster the generalizability of the feature representation model.
3) The confounding factors in the pseudo-label-driven self-training process are analyzed from a causal inference perspective. A de-confounding framework is developed based on causal intervention, which introduces the backdoor adjustment to disentangle the negative effects caused by the confounding bias of noisy pseudo-labels.

The rest of this article is organized as follows. We begin by reviewing related work in the field, as described in Section II. We then identify potential issues through theoretical analysis and experimental validation, which form the basis of our investigation, as detailed in Section III. After that, Section IV gives the detail of our proposed method CausalCD. To verify the effectiveness of CausalCD, we conducted corresponding experiments, which are presented in Section V. Finally, we conclude this article in Section VI.

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II. Related Work

A. Self-Supervised SAR Image Change Detection

Self-supervised SAR image change detection does not require labeling of changed or unchanged training samples, and utilizes certain attributes or structure of data as *pseudo-labels* for model learning. Gao et al. [17] pioneered a framework for self-supervised SAR image change detection that utilizes selected changed and unchanged pixels as pseudo-labels for training principal component analysis network (PCANet) through Gabor wavelets and fuzzy C-means (FCM). Their method has achieved significant results compared to the traditional algorithms. Since then, many pseudo-label-driveb self-supervised SAR change detection algorithms have been developed. In Ma et al.'s work [18], the change detection network is trained using the pseudo-labels generated by clustering, and complementary features are used to integrate the spatial and frequency domain features, so as to improve the change detection performance. Zhang et al. [19] proposed a hybrid convolutional and self-attention network called convolution and attention mixer (CAMixer), which also employs a hierarchical FCM to generate pseudo-labels to train the network.

However, the clustering-based pseudo-labels have two drawbacks: 1) mis-classification in the generated pseudo-labels and 2) an imbalance between changed and unchanged pseudo-labels. To address the above issues, many studies have been carried out, with corresponding solutions proposed. For example, Li et al. [20] combined salient region detection with the PCANet to alleviate the problem of excessive training samples. They further refined the selection of candidate samples by implementing a uniform selection strategy to counteract the uneven distribution encountered in the generation of pseudo-labels. Peng et al. [21] proposed an improved FCM algorithm, multi-hierarchical fuzzy local information C-means clustering (MH-FLICM), which can achieve accurate detection of the boundary regions and accurate identification of changed and unchanged areas with high confidence. Meng et al. [22] presented a joint cross entropy and mean absolute error noise-tolerant loss for the incorrect pseudo-labels generated by clustering, which can improve the robustness of the change detection model to noisy labels. Zhang et al. [23] introduced a clustering regularization term based on c-mean on the basis of fine-tuning loss to address the errors in the generated pseudo-labels, so as to learn discriminative and compact features and reduce the influence of selection criteria. Different from the above improved algorithms, we address the problem of noisy pseudo-labels in the self-supervised change detection framework from a different perspective, namely, that of causal inference.

B. Causal Inference

Causal inference is a statistical tool for modeling the intrinsic connections between variables. According to Yoshua, winner of the Turing Award, machine learning algorithms should recognize causal relationships to achieve powerful AI [24]. Different from deep learning algorithms, causal learning can provide incomparable interpretability, which enables researchers to gain a deeper understanding of the research subjects [25]. Due to the impressive performance of causal inference theory, many researchers have begun to utilize causal relationships to design new methods for SAR image interpretation. The causal inference theory can be utilized to determine the causal relationship between key factors in the current SAR image interpretation field, eliminate certain spurious confounding factors introduced in the process of image interpretation, or provide new perspectives for rethinking specific learning problems. For example, Guo et al. [26] investigated the neural network representation of SAR images from the perspective of causal relationship, and proposed a causal adversarial autoencoder for SAR target image representation and recognition with limited training samples. Wang et al. [27] proposed a causal intervening automatic target detection (CIATR) method to solve the problem of limited SAR data. Li et al. [28] analyzed the non-causal relationship of deep learning in SAR automatic target detection based on the Shapley value, and the factors affecting the over-fitting of clutter were clearly demonstrated by intervening in the comparability of clutter and model parameters, thus reducing the errors in training. Wang et al. [29] constructed a causal SAR automatic target detection model based on pairwise invariance, which not only solved the critical problem caused by the limited SAR data but also deduced an effective principle solution and achieved better performance.

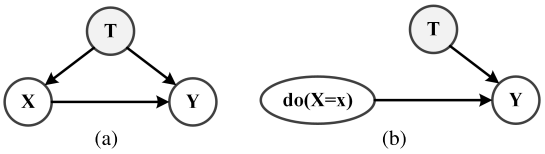


Fig. 2. Illustration of the causal intervention. (a) Common SCM for causal inference. (b) Causal intervention performed on (a).

Causal intervention is the basis to define the causal effects. In SCM, the intervention is represented by the *do* operator. Fig. 2(a) shows a common SCM for causal inference, in which the three variables $\mathbf{T}$, $\mathbf{X}$, and $\mathbf{Y}$ represent the confounding variable, treatment variable, and outcome variable, respectively. In Fig. 2(b), an intervention is performed on the treatment variable $\mathbf{X}$, i.e., $\text{do}(\mathbf{X} = \mathbf{x})$. The intervened variable $\mathbf{X}$ is no longer affected by its parent variable $\mathbf{T}$ and does not have any more directed edges into it. In other words, when $\mathbf{X}$ is intervened, $\mathbf{T}$ no longer affects both the treatment variable $\mathbf{X}$ and the outcome variable $\mathbf{Y}$, at which point $\mathbf{T}$ is no longer a confounding variable. Suppose the spurious correlation caused by the confounding variables in the training data is stronger than the actual causal relationship. In that case, the spurious correlation may be learned by machine learning models and used for prediction. In addition, the spurious correlation that holds in the training set may not be valid in the test set, thus causing various issues such as generalization and interpretability of the machine learning model. Therefore, on the basis of the above works, we construct a specific SCM for self-supervised SAR image change detection and investigate the confounding effect caused by noisy pseudo-labels from the causal perspective.

C. Graph Contrastive Learning

Recently, GCL has attracted significant attention and has established itself as a potent learning paradigm within the

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domain of unsupervised graph representation learning. GCL can improve network training when there are limited labeled data. Inspired by the contrastive learning, GCL utilizes the advanced representation learning ability of graph convolutional network (GCN) and extracts the high-quality graph embedding of input augmented graphs by comparing the similarities and differences between augmented graphs, so as to achieve comprehensive generalizability in various downstream tasks by maximizing mutual information [16], [30].

On the basis of pioneering works, many GCL-based remote sensing image processing algorithms have been proposed recently. For example, Hu et al. [31] introduced I2M-GAN, a novel graph attention network that integrates both intra-modal and inter-modal contrastive learning for aligning SAR and optical images. Xiao et al. [32] put forward the knowledge-guided contrastive domain adaptation (K-CDA) approach. This method aims to tackle the challenges of unshared and unbalanced unsupervised domain adaptation in aerospace data. Li et al. [33] utilized GCL to pre-train the SAR image classification model with a wealth of unlabeled images at their disposal. They also introduced two innovative strategies for the construction of positive sample pairs, thereby enhancing the feature learning process for contrasting SAR images. Liu et al. [34] presented a dual dynamic GCN for SAR scene classification using contrastive learning, designed a new contrastive loss for structural understanding, enhancing GCN node representation and communication. We re-examine the GCL framework from the perspective of causal inference, based on which a new causal graph data augmentation strategy is proposed. The graph structure diversity and semantic diversity are introduced as constraints to ensure the generation of hard positive and negative instances of the original graphs, thereby improving the generalization and rationality of the graph feature representation model.

III. Motivation

Inspired by the above works, there are two questions about key steps of the pipeline shown in Fig. 1.

1) In stage 1, can the GCL-based model ensure the generation of appropriate and generalized feature representations for pixels in bitemporal SAR images?
2) In stage 2, will the incorrect samples in the pseudo-label set generated by clustering have a negative impact on the change detection model?

To answer the above questions, we perform a more in-depth analysis of the self-supervised SAR change detection pipeline both theoretically and experimentally first.

A. Fragile Graph Contrastive Learning Model

The GCL model requires generating positive and negative instances for the original graph, after which the similarities and differences between the graph data are learned to make positive instances get close to each other and negative instances get away from each other in the feature space, so as to achieve the generalizability of the downstream tasks by maximizing the mutual information. In most related works, the positive instances of the original graph are generated by graph data augmentation, including random deleting edges or nodes, and random masking node features [16], as shown in Fig. 3. However, there is no theoretical study that explains why this data augmentation strategy provides the best model representation. In addition, when randomly deleting edges or nodes, irrelevant information may be captured.

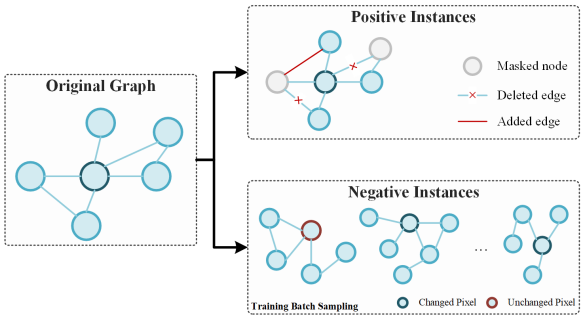


Fig. 3. Illustration of the traditional graph data augmentation [16]. The positive instances of the original graph are generated by randomly deleting edges or nodes, and random masks of node features. The negative instances are selected from the graph structure constructed from other pixels, such as other graph structures in the same training batch.

On the other hand, the negative instances of the original graph are selected from the graph structure constructed from other pixels, such as other graph structures in the same training batch. However, the pixels in the change detection tasks are categorized into two classes (changed class and unchanged class), so there is a high possibility that the randomly selected negative instances belong to the same class as the original graph (e.g., the center pixel of the original graph in Fig. 3 is a changed class pixel (blue one), while there is a graph of unchanged class (red one) sampled from the training batch). Therefore, it is likely that the traditional graph data augmentation method will negatively affect the model learning, causing the model to learn irrelevant features and resulting in a significant degradation of the model performance.

B. Noisy Pseudo-Label Self-Training Dataset

In the process of generating pseudo-labels, the presence of incorrect samples is inevitable, and training sets with incorrect pseudo-labels can lead to a significant decrease in the performance of the change detection models. Therefore, mitigating the negative impact of noisy pseudo-label sets on model performance is an important issue in the field of self-supervised SAR change detection.

IV. Proposed CausalCD

To address the two problems in the pipeline (please refer to Section III for specific details), a principled and fundamental solution is proposed from the perspective of causal inference. First, the process of self-supervised SAR image change detection is formalized based on the causal inference theory, and the problems in change detection are analyzed from the perspective of causal inference, as described in Section IV-D1. After that, to tackle the first problem of fragile GCL, we design a novel graph data augmentation strategy based on the causal invariant mechanism, as described in Section IV-D2. For the second problem of noisy pseudo-label self-training datasets, we de-confound the pseudo-labels

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based on causal interventions to eliminate the interference of incorrect samples on the model, as described in Section IV-D3. The comprehensive architecture of CausalCD is depicted in Fig. 4.

Preliminaries: Let us start by introducing how to construct a graph structure for any arbitrary pixel v_i, as shown in Fig. 5. Given two registrated SAR images I_1, I_2 of the same area, with temporal phases T_1, T_2. First, we perform superpixel segmentation on I_1 and I_2 based on the simple linear iterative clustering (SLIC) algorithm [35] and obtain two sets of homogeneous and compact superpixel patches, denoted as $SP_1 = \{sp_1^1, sp_2^1, \ldots, sp_{Ns_1}^1\}$, $SP_2 = \{sp_1^2, sp_2^2, \ldots, sp_{Ns_2}^2\}$, respectively. Ns_1 and Ns_2 represent the superpixel size. Then, we merge the two sets of superpixel segmentation results to obtain the superpixel set $SP = SP_1 \cup SP_2 = \{sp_1, sp_2, \ldots, sp_{Ns}\}$. It is important to note that to guarantee that each pixel can obtain k nearest values within its superpixel patch in subsequent steps, we set a threshold to merge smaller superpixel patches into the surrounding ones.

For the graph structure $\mathcal{G}_i$ of the pixel v_i, it is composed of nodes V_i, edges ε_i, the adjacency matrix A_i, and the feature matrix F_i

$$\mathcal{G}_i = (V_i, \varepsilon_i, A_i, F_i). \tag{1}$$

A. Graph Nodes

The initial feature f_i of pixel v_i consists of its bitemporal magnitude features

$$f_i = \left(f_i^{\text{mag}-T_1}, f_i^{\text{mag}-T_2}\right). \tag{2}$$

Next, we obtain the corresponding k-neighbor pointset V_i of pixel v_i within the corresponding superpixel patches as graph nodes based on the similarity function

$$\text{Sim}(v_i, v_j) = \left(\left\|f_i - f_j\right\|_2\right)^2 \tag{3}$$

where $v_i, v_j \in sp_n$. Therefore, $V_i = \{v_i \cup v_j, j = 1, 2, \ldots, k\}$ is the graph nodes.

B. Adjacency Matrix

The adjacency matrix represents a matrix of the connectivity between nodes within a graph. In an adjacency matrix, the rows and columns represent the nodes in the graph, and the elements of the matrix indicate the state of connection between the nodes. We define the adjacency matrix A_i of pixel v_i as

$$A_i = e^{-\left(\frac{C_d}{\sigma(C_d)}\right)^2 - \left(\frac{F_d}{\sigma(F_d)}\right)^2} \tag{4}$$

where $C_d(p, q) = \|c_p - c_q\|^2$ is the Euclidean distance between node coordinate c_p and c_q. $F_d(p, q) = \|f_p - f_q\|^2$ is the feature distance between node v_p and node v_q. The function $\sigma(\cdot)$ represents the operation of taking the average of the elements in each row of a matrix. Then, convert matrix A_i into a symmetric matrix and set each element to either 0 or 1

$$\hat{A}_i = \begin{cases} 1, & \text{sigmoid}\left(\dfrac{A_i + A_i{}^T}{2}\right) \geq \alpha_0 \\ 0, & \text{sigmoid}\left(\dfrac{A_i + A_i{}^T}{2}\right) < \alpha_0 \end{cases} \tag{5}$$

where $\hat{A}_i \in \{0, 1\}^{(k+1)\times(k+1)}$. α_0 is a threshold determining whether to set the entry of $\hat{A}_i$ as 1 or 0.

C. Graph Edges

If the element $\hat{A}_{pq}$ in matrix $\hat{A}_i$ equals 1, then there is an edge between node v_p and node v_q; if it equals 0, then no edge exists. Therefore, the graph edges can formulated as

$$\varepsilon_i = \left\{e_{pq} = \left(v_p, v_q\right), \hat{A}_{pq} = 1 \text{ and } p \neq q\right\}. \tag{6}$$

D. Feature Matrix

$F_i = \{f_i \cup f_j, j = 1, 2, \ldots, k\} \in \mathbb{R}^{(k+1)\times D_0}$ is the matrix consists of the features of all nodes. D_0 is the dimension of f_i.

Therefore, we obtain the graph structure for each pixel, and fed these graphs as the basic processing unit to feature extraction encoders.

1) Structure Causal Model Construction and Analysis: Given a bitemporal SAR dataset, the graph structures $\mathcal{G}$ are constructed pixel by pixel, which serves as the basic processing unit of the network. We define the generated noisy pseudo-label set as $L = E \cup T = \{l_i | l_i = c, c \in \mathcal{C} = \{0, 1\}\}_{i=1}^{N_L}$. L consists of the correct label set E, and the incorrect label set T. N_L is the total number of samples in the pseudo-label set L. If l_i is 0, it represents the changed sample. Otherwise, it is the unchanged sample. Therefore, we define the overall objective of self-supervised change detection as

$$\mathcal{F} = \underset{\Theta}{\text{argmin}} \sum_{\mathcal{G}\in\mathbb{G}} \mathcal{L}(\delta(H(R(\mathcal{G}))), l) \tag{7}$$

where our overall objective is to learn the change detection network $\mathcal{F}$, which can be formalized into three steps

$$\begin{aligned} x &= R(\mathcal{G}), R(\cdot; \Theta_R) : \mathcal{G} \mapsto \mathbb{R}^{D_r} \\ h &= H(x), H(\cdot; \Theta_H) : \mathbb{R}^{D_r} \mapsto \mathbb{R}^{D_h} \\ z &= \delta(h), \delta(\cdot; \Theta_\delta) : \mathbb{R}^{D_h} \mapsto \mathbb{R}^2 \end{aligned} \tag{8}$$

where the graph feature extraction network $R(\cdot; \Theta_R) : \mathcal{G} \mapsto \mathbb{R}^{D_r}$ transforms the graph structures $\mathcal{G}$ into feature representations x. The change detection network $H(\cdot; \Theta_H) : \mathbb{R}^{D_r} \mapsto \mathbb{R}^{D_h}$ maps x into feature h, and the change detection regression layer predicts the score z of h as a changed pixel. $\delta(\cdot; \Theta_\delta) : \mathbb{R}^{D_h} \mapsto \mathbb{R}^2$ is logsoftmax. The overall hyperparameters $\Theta = \{\Theta_R, \Theta_H, \Theta_\delta\}$ are optimized using cross-entropy loss $\mathcal{L}$.

Based on the causal inference theory, we construct an SCM to describe the whole change detection framework, as illustrated in Fig. 6. The self-supervised change detection framework mainly includes two components: graph feature representation and pseudo-label-driven self-supervised training. Next, the SCM is described in detail.

a) SCM—graph feature representation: When performing the downstream change detection task on a graph, the topological relationships between nodes and edges, as well as the features of graph, are defined as the graph structural information. The class and attributes of the graph represent its semantic information. Given that the semantic information aptly characterizes the graph, we focus our objective for the graph class prediction on accurately estimating the graph's semantic properties.

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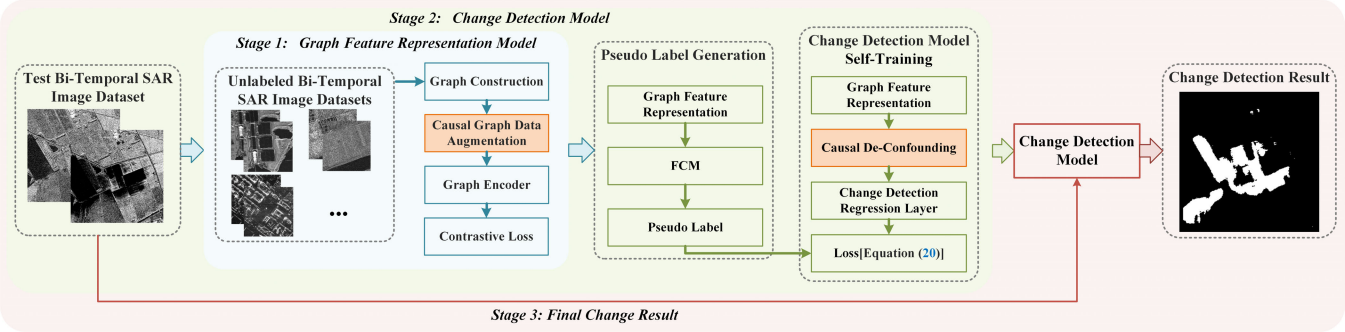


Fig. 4. Comprehensive architecture of CausalCD. Compared to Fig. 1, there are two improvements: First, a novel causal graph data augmentation strategy based on the causal invariant mechanism is designed to ensure the generation of hard positive and negative instances of the original graph. Second, Causal De-Confounding is adopted to eliminate the interference of noisy pseudo-labels on the model.

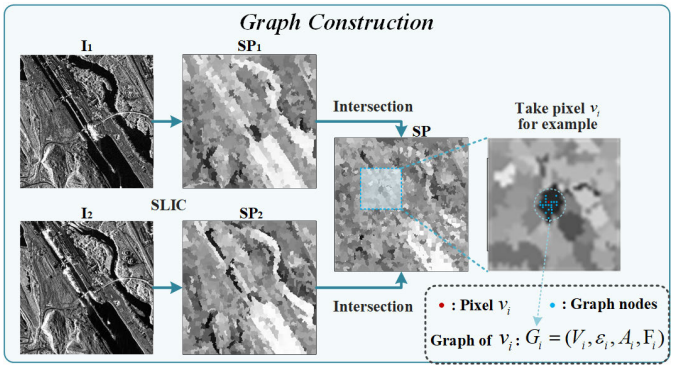


Fig. 5. Illustration of the graph construction for any arbitrary pixel v_i.

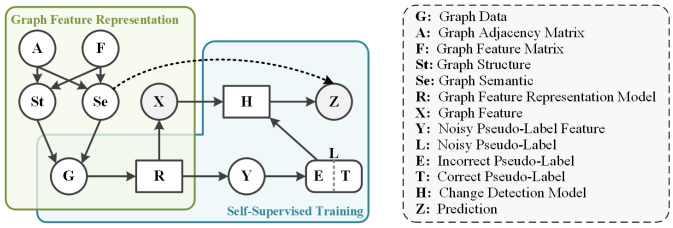


Fig. 6. Green part signifies the graph feature representation model, and the blue part indicates pseudo-label generation-based self-training.

We construct an SCM based on the following three assumptions: 1) the graph is generated from structure and semantic variables; 2) there is a certain correlation between structure and semantics; and 3) the downstream change detection task is semantically related, meaning that the semantics of the graph structure corresponding to a pixel determine whether it belongs to the changed class or unchanged class. Therefore, we construct an SCM in accordance with Pearl's graphical mode [25] to formalize the above definitions and assumptions, as depicted in green in Fig. 6. **G** denotes the input graph data. **R** is the graph feature representation model. **X** represents the graph features. **A** and **F** are the adjacency matrix and feature matrix of graph. **St** is the potential variable describing the structure of graph, which is described by the node connectivity (described by the adjacency matrix **A**) and node features (described by the feature matrix **F**). **Se** is the potential variable describing the semantics of graph, which is described by the class probability distribution of the entire graph. The graph is encoded using a graph encoder to obtain its probability distribution representation: $P(\mathbf{G}) = \text{GraphEncoder}(\mathbf{A}, \mathbf{F})$. **Z** is the prediction result of the downstream change detection task. The causal correlations among these variables in the SCM are as follows.

1) The causal link $\mathbf{A} \rightarrow \mathbf{St}$, $\mathbf{A} \rightarrow \mathbf{Se}$, $\mathbf{F} \rightarrow \mathbf{St}$, and $\mathbf{F} \rightarrow \mathbf{Se}$ indicate that both **St** and **Se** are determined by the adjacency matrix and the feature matrix, and there is a potential correlation between **St** and **Se**.
2) The causal link $\mathbf{St} \rightarrow \mathbf{G}$ and $\mathbf{Se} \rightarrow \mathbf{G}$ indicate that **G** is generated based on structure and semantics.
3) The causal link $\mathbf{Se} \rightarrow \mathbf{Z}$ indicates that the semantics directly influence the change detection task. Meanwhile, the lack of a connection between **St** and **Z** signifies that the structure has no direct effect on the downstream change detection task.
4) The causal link $\mathbf{G} \rightarrow \mathbf{R} \rightarrow \mathbf{X}$ represents the graph feature representation process, where the feature extractor uses graph to pre-train a representation model $R(\cdot)$ with unlabeled samples, i.e., $x = R(\mathcal{G})$ in (7). This is done to provide a robust graph feature representation for the downstream change detection task.

b) SCM—pseudo-label-driven self-supervised training: The training set of the change detection model is obtained using a heuristic pseudo-label generation algorithm, which has been proposed in numerous studies for self-supervised change detection. To ensure a fair comparison with other algorithms, we adopt the simplest pseudo-label generation method, i.e., FCM [36]. By clustering the difference image of the test dataset into three classes and sorting the three clusters by their mean values. The cluster with the highest mean pixel value is set as the changed class, while the one with the lowest mean is assigned as the unchanged class. The SCM of the self-supervised change detection training process based on pseudo-label generation is shown in blue in Fig. 6, in which **Y** is the noisy pseudo-label feature, **L** is the noisy pseudo-label, and **H** is the change detection model. The causal correlations among these individual variables in the SCM are as follows.

1) The causal link $\mathbf{R} \rightarrow \mathbf{Y} \rightarrow \mathbf{L}$ represents the pseudo-label generation process.
2) The causal link $\mathbf{X} \rightarrow \mathbf{H} \rightarrow \mathbf{Z}$ denotes the change prediction process, $y = \delta(h = H(x = R(\mathcal{G})))$ in (7).
3) The causal link $\mathbf{X} \rightarrow \mathbf{H} \leftarrow \mathbf{L}$ represents $\mathcal{L}(y, l)$ in (7).

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2) Graph Contrastive Learning With Causal Invariant Mechanism: The training of deep networks typically depends on a substantial amount of supervised signals, which can be costly to acquire, especially for SAR images. However, there are abundant unlabeled samples available. Consequently, utilizing pre-trained representations for change detection tasks can diminish the learning costs, as these do not require labeled training samples. To obtain the pre-trained feature representations, we employ an SSL model that leverages a GCL-based framework, specifically designed to extract feature representations from graph data. However, according to the analysis of the problems in the key step in GCL (i.e., graph data augmentation), as discussed in Section III, it reveals that traditional graph data augmentation methods are prone to generating false positive and negative instances of the original graph. Therefore, we reconsider the graph data augmentation process from a causal perspective first and design a new graph data augmentation strategy. This strategy aims to generate hard positive and negative instances, employing constraints based on graph structure diversity and semantic diversity.

Utilizing the causal invariant mechanism and GCL strategy, it can be concluded that intervening on **St** does not change the conditional distribution within our constructed SCM, which can be expressed as

$$P^{\mathrm{do}(\mathbf{St}=\mathrm{st}_i)}(\mathbf{Z}|\mathbf{Se}) = P^{\mathrm{do}(\mathbf{St}=\mathrm{st}_j)}(\mathbf{Z}|\mathbf{Se}) \tag{9}$$

where $P^{\mathrm{do}(\mathbf{St}=\mathrm{st})}$ denotes the intervention of the graph structure, we consider **Se** as a representation that stably predicts change under distribution, i.e., the semantic variable **Se** is a robust representation for the change detection task. Therefore, we can employ the graph data augmentation method that preserves the semantic content of the graph structure. Specifically, an intervention is implemented on the structure variable **St** via data augmentation. Although we cannot generate all possible instances for fixed graph data through graph data augmentation, we can generate a large number of different structural data as an augmentation to learn better representations. Further, we can conclude that the feature representation model $\mathbf{R}(\mathbf{X})$ satisfies the following invariant prediction:

$$P^{\mathrm{do}(a_i)}(\mathbf{Z}|\mathbf{R}(\mathbf{X})) = P^{\mathrm{do}(a_j)}(\mathbf{Z}|\mathbf{R}(\mathbf{X})) \quad \forall a_i, a_j \in \mathcal{A} \tag{10}$$

where $\mathcal{A} = \{a_1, a_2, \ldots, a_{na}\}$ is a set of graph data augmentation which simulate interventions on **St**, and $P^{\mathrm{do}(a_i)}$ is equivalent to $P^{\mathrm{do}(\mathbf{St}=\mathrm{st}_i)}$ in (9).

Based on the above analysis, we define the hard positive and negative instances generated from the original graph data from a causal perspective. The hard positive instances should satisfy: *the semantics remain unchanged despite significant structural disturbances.* The hard negative instances should satisfy: *the semantics change even with minor structural disturbances.* Unlike the random graph data augmentation approaches in other works, we describe the structural perturbation and feature masking during graph data augmentation by defining two learnable matrices M_s and M_f. The generation of hard positive and negative instances of the original graph is guaranteed via learning. Specifically, for the original graph $\mathcal{G} = \{A, F\}$ with its adjacency matrix denoted by A, the matrix is transformed after structural perturbation

$$\hat{A} = \begin{cases} 1, & \mathrm{sigmoid}\left(\dfrac{M_s \times A + (M_s \times A)^T}{2}\right) \geq \alpha_1 \\ 0, & \mathrm{sigmoid}\left(\dfrac{M_s \times A + (M_s \times A)^T}{2}\right) < \alpha_1 \end{cases} \tag{11}$$

where $M_s \in \mathbb{R}^{(k+1)\times(k+1)}$ is a learnable matrix, we can use M_s to simulate all structural perturbation types, including dropping and adding edges or nodes. α_1 is a threshold determining whether to set the entry as 1 or 0. Note that we adopt matrix multiplication here instead of taking the Hadamard product since it cannot add an edge to the adjacency matrix.

The feature mask matrix M_f can mask the feature matrix of the original graph data so that the model can obtain the key information in the features. Therefore, after the feature masking, the feature matrix of the original graph data is transformed from F to

$$\hat{F} = \hat{M}_f \circ F \tag{12}$$

where $\circ$ denotes Hadamard product. M_f is a learnable feature mask matrix. We need to ensure its definition domain belongs to 0,1, so we have

$$\hat{M}_f = \begin{cases} 1, & \mathrm{sigmoid}(M_f) \geq \alpha_2 \\ 0, & \mathrm{sigmoid}(M_f) < \alpha_2 \end{cases} \tag{13}$$

where α_2 is a threshold determining whether to set the entry as 1 or 0. Therefore, we obtain the augmented graph $\mathcal{G}'$ after graph data augmentation

$$\mathcal{G}' = \{\hat{A}, \hat{F}\}. \tag{14}$$

To describe the structural relationship between the generated augmented instances and the original graph data, we define the *graph structure diversity* D_{st} based on the adjacent matrix and the feature matrix

$$D_{\mathrm{st}} = \left\|A - \hat{A}\right\|_1 - \left\|\hat{M}_f\right\|_1 \tag{15}$$

where $\|\cdot\|_1$ denotes the bi-paradigm, which is used to measure the similarity between the matrices. A larger $\|\hat{M}_f\|_1$ indicates that a smaller portion of features is masked. Therefore, the similarity between F and $\hat{F}$ will be high. Therefore, the smaller the value of D_{st}, the more similar the structure is between the original and augmented graph. Furthermore, to describe the semantic relationship between the generated instances and the original graph data, we define the *graph semantic diversity* D_{se} based on the class probability distribution

$$D_{\mathrm{se}} = d_{\mathrm{KL}}\left(P(\mathcal{G}), P\left(\mathcal{G}'\right)\right) \tag{16}$$

where d_{KL} is the Kullback–Leibler divergence, which measures the similarity between two probability distributions [37]. $P(\cdot)$ represents the probabilities of the classes to which the graph belongs. Similarly, the smaller the value of D_{se}, the more similar the semantics between the original graph and the augmented graph. Thus, the loss functions of generated hard positive and negative instances are defined as

$$\begin{aligned} \mathcal{L}_{\mathrm{pos}} &= e^{\frac{1}{D_{\mathrm{st}} - D_{\mathrm{se}}}} \\ \mathcal{L}_{\mathrm{neg}} &= e^{D_{\mathrm{st}} - D_{\mathrm{se}}}. \end{aligned} \tag{17}$$

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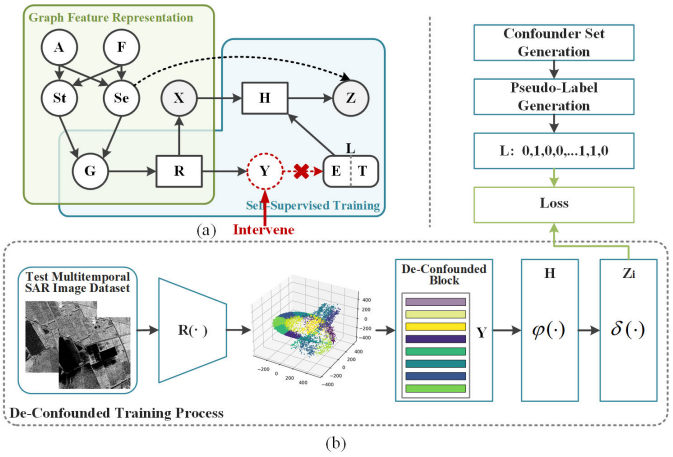


Fig. 7. Overview of the de-confounded process defined in (23). (a) Proposed de-confounded causal graph with causal intervention. (b) Detailed implementation of de-confounded.

Therefore, we construct hard positive instance and hard negative instance generation networks separately, with the objective of minimizing the above loss functions. During pre-training, a training batch of graphs is randomly sampled and processed through contrastive learning. With the causal graph data augmentation proposed above, we can generate hard positive instance and hard negative instances for the original graph data. Then, we get the corresponding features $Z_n, Z_{n_+}^+, Z_{n_-}^-$ of these graph instances for the nth graph in the training batch. Therefore, we define the overall goal of the graph feature representation model as follows:

$$\mathcal{L}_n = -\log \frac{e^{\left(Z_n^T Z_{n_+}^+/\tau\right)}}{e^{\left(Z_n^T Z_{n_+}^+/\tau\right)} + e^{\left(Z_n^T Z_{n_-}^-/\tau\right)}} \tag{18}$$

where τ is the temperature hyperparameter, and the analysis of the impact of τ size is in Section V-H.

3) De-Confounded Self-Training With Causal Intervention: According to the causal inference theory [25], $\mathbf{X}$ and $\mathbf{Z}$ appear spurious correlation due to the confounding introduced by the backdoor path $\mathbf{X} \leftarrow \mathbf{R} \rightarrow \mathbf{Y} \rightarrow \mathbf{L} \rightarrow \mathbf{H} \rightarrow \mathbf{Z}$. As a result, model $\mathcal{F}$ in (7) cannot represent the direct causal correlation between $\mathbf{X}$ and $\mathbf{Z}$. The correct pseudo-label set $\mathbf{E}$ within $\mathbf{L}$ assists in better learning the change detection-specific representation space through $\mathbf{X} \rightarrow \mathbf{H} \rightarrow \mathbf{Z}$. Conversely, the incorrect pseudo-label set $\mathbf{T}$ distorts the representation space through the backdoor path, thus negatively affecting the learning of the change detection model to some extent. Therefore, this inspires us to design an intervening causal graph to eliminate the confounding bias caused by the backdoor path on the causal correlation between $\mathbf{X}$ and $\mathbf{Z}$, so as to find out the real causal correlation between $\mathbf{X}$ and $\mathbf{Z}$.

The adjusted intervention SCM is shown in Fig. 7. It can be seen that by blocking the causal link of $\mathbf{X} \leftarrow \mathbf{R} \rightarrow \mathbf{Y} \rightarrow \mathbf{L} \rightarrow \mathbf{H} \rightarrow \mathbf{Z}$, the confounding path is blocked so that there is no spurious correlation between the pseudo-label generation process and the model learning. There will be a direct causal effect between $\mathbf{X}$ and $\mathbf{Z}$. Moreover, because the feature representation model $\mathbf{R}$ and the pseudo-label $\mathbf{L}$ are intractable, we choose $\mathbf{Y}$ for stratifying to achieve backdoor adjustment. The main idea of causal intervention is: 1) cutting off the $\mathbf{Y} \rightarrow \mathbf{L}$ link in the backdoor path and 2) stratifying $\mathbf{Y}$ into pieces to construct a confounder set $\hat{Y} = \{y_i\}_{i=1}^{N_C}$, $y_i \in \mathbb{R}^{D_r}$. N_C is a hyperparameter representing the size of the confounder set $\hat{Y}$. The effect of N_C value on model performance is discussed in Section V-F. The adjustment formula for eliminating the confounding effect is denoted as

$$P(\mathbf{Z}|\text{do}(\mathbf{X})) = \sum_{y_i} P\left(\mathbf{Z}\middle|\mathbf{X}, \mathbf{H}, \hat{Y} = y_i\right) P(y_i). \tag{19}$$

How can we stratify the confounder set $\hat{Y}$? Given the large number of samples in the pseudo-label set generated by clustering, we employ K-means++ with PCA to learn the confounder set $\hat{Y}$, with y_i representing the central feature of each cluster. However, performing network forward passes for all N_C clusters, as required by (19), is computationally expensive. To mitigate this, we apply the normalized weighted geometric mean (NWGM) [38] to move the external summation to the feature level, thereby reducing computational overhead. Consequently, we define the new adjustment formula as follows:

$$\begin{aligned} P(\mathbf{Z}|\text{do}(\mathbf{X})) &= \sum_{y_i} P(\mathbf{Z}|\mathbf{X}, \mathbf{H} = g(\mathbf{X}, y_i)) P(y_i) \\ &\approx P\left(\mathbf{Z}\middle|\mathbf{X}, \hat{h} = \sum_{y_i} g(\mathbf{X}, y_i) P(y_i)\right). \end{aligned} \tag{20}$$

By obstructing the backdoor pathway, $\mathbf{X}$ is afforded an equitable chance to include each y_i into the prediction of $\mathbf{Z}$, conditional upon a prior $P(y_i)$. We define $g(\cdot)$ as the soft attention to normalize similarity between x and the confounder set $\hat{Y}$

$$g\left(x, \hat{Y}\right) = \text{Attention}\left(x, \hat{Y}\right) = \delta\left(\frac{(\mathbf{W_1}x)^T\left(\mathbf{W_2}\hat{Y}\right)}{\sqrt{N_C}}\right)\hat{Y} \tag{21}$$

where δ is logsoftmax. $\mathbf{W_1}$, $\mathbf{W_2} \in \mathbb{R}^{N_C \times D_r}$ represent two trainable projection matrices employed to map x and $\hat{Y}$ onto a shared space. $(N_C)^{1/2}$ is a constant factor for feature normalization. Therefore, we define the adjustment feature

$$\begin{aligned} \hat{h} &= \sum_{i=1}^{N_C} g_i\left(x, \hat{Y}\right) P(y_i) \\ &= \sum_{i=1}^{N_C} \delta\left(\frac{(\mathbf{W_1}x)^T(\mathbf{W_2}y_i)}{\sqrt{N_C}}\right) \cdot y_i \cdot \frac{|y_i|}{\sum_i |y_i|} \end{aligned} \tag{22}$$

where δ is logsoftmax. Finally, compared to the baseline model $\mathcal{F}$ in (7), the final change prediction $\hat{\mathcal{F}}$ is achieved by the following formula:

$$\hat{\mathcal{F}} = \underset{\Theta'}{\text{argmin}} \sum_{\mathcal{G} \in \mathbb{G}} \mathcal{L}\left(\delta\left(\left[R(\mathcal{G}), \hat{h}\right]\right), l\right) \tag{23}$$

where $[\cdot, \cdot]$ is the concatenation operator. Our improved overall objective is to learn the change detection network $\hat{\mathcal{F}}$. $\mathcal{L}$ is cross-entropy loss. The overall hyperparameters $\Theta' = \{\Theta'_R, \Theta'_H, \Theta'_\delta\}$ are optimized using cross-entropy loss $\mathcal{L}$.

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TABLE I
INTRODUCTION OF THREE SAR IMAGE SCENES

No.	Area	Resolution(m)	Size(pixels)	Sensor	Band	Track	Polarization	Acquisition Dates	Change Type
1	Conselice	0.57	1500 × 1500	Umbra	X	Ascending	VV	2023/6/6 2023/6/7	Flood
2	Rotterdam	0.5	3000 × 2000	ICEYE	X	Ascending	VV	2021/4/5 2021/4/7	Ship targets
3	Panama Canal	0.5	500 × 500	TerraSAR	X	Ascending	HH	2013/9/26 2013/12/12	Ship targets and land

V. EXPERIMENTS

In this section, we begin by introducing three bitemporal SAR scenes that we used in our study, and provide details about the parameter configurations set in our experiments. Subsequently, we compare the results of our proposed CausalCD with those of previous state-of-the-art methods from qualitative and quantitative perspectives. Finally, we conduct several ablation experiments to validate the effectiveness of the proposed CausalCD.

A. Dataset Descriptions and Implementation Details

To verify the effectiveness of CausalCD proposed in this article, we applied it to three bitemporal registered SAR image scenes. Table I summarizes the characteristics of these SAR image scenes, detailing their coverage areas, resolutions, size, sensors, tracks, polarization modes, acquisition dates, and change types. The first scene includes SAR images of Conselice in Italy, acquired by Umbra satellite sensor in June 2023, depicting the impact of flood disasters and the changes in the submerged agricultural lands. The second SAR image scene was captured from the port area of Rotterdam in the Netherlands by ICEYE satellite sensor, revealing significant changes in the number and positions of ship targets. The last scene presents SAR images of the Panama Canal region, captured by TerraSAR satellite sensor in 2013, and includes changes in ships and land areas.

CausalCD is implemented using the PyTorch framework and runs on a desktop equipped with an NVIDIA RTX 2080Ti GPU, which has 11 GB of memory. The configuration of the parameters is as follows: the initial learning rate is set to 0.001, and the batch size is set to 32. The model employs adaptive moment estimation with weight decay (AdamW) [39] as its optimizer. Initialization for variables and parameters in the CausalCD is set as $\alpha_0 = 0.1, \alpha_1 = 0.2, \alpha_2 = 0.2, \tau = 0.1$. Parameter values remain fixed during the experiments except for parametric analysis. Section V-H offers an in-depth examination of parameter sensitivity.

B. Compared Methods and Evaluation Metrics

To evaluate the superiority of the CausalCD, five unsupervised methods are used as compared methods.

1) *PCANet [17]:* A pseudo-label generation strategy is designed based on Gabor wavelets and FCM, and a change detection model is obtained by training PCANet with pseudo-labels.
2) *Deep change vector analysis (DCVA) [40]:* An unsupervised change detection algorithm that uses a pre-trained CNN to generate deep change vectors, which are then converted to binary vectors for identifying changed pixels. The algorithm analyzes the binary vectors to detect changes in the image, considering the spatial context of the pixels.
3) *Convolutional-wavelet neural network (CWNN) [41]:* A change detection approach that merges dual-tree complex wavelet transform with CNNs for classification, reducing speckle noise. It also introduces a virtual sample generation for CWNN training to overcome limited samples.
4) *Improved non-local patch-based graph (INLPG) [42]:* An unsupervised change detection algorithm, which assumes that heterogeneous images share the same structural information, and then uses a graph to measure the structural consistency between heterogeneous images.
5) *Iterative robust graph and Markovian co-segmentation method (IRG-McS) [43]:* An object-based unsupervised change detection algorithm, which utilizes the inherent structural consistency between heterogeneous images to construct robust graphs representing the images' structures, and then compares the degree of change through graph mapping.
6) *Feature-mutual-representation-based graph domain adaptive network (FGDANet) [44]:* An unsupervised graph domain adaptation network, comprising a pseudo-siamese backbone formed by two custom unsupervised learning domains. Through a graph domain adaptation strategy, it fully utilizes information from different domains to enhance feature representation and generalization capabilities.
7) *Multiscale self-attention deep clustering (MSDC-SAM) [45]:* Combining unsupervised clustering with CNNs for end-to-end feature learning and cluster optimization, the approach incorporates k-means++, octave convolution, and self-attention mechanisms to enhance noise resistance and capture spatial information, while employing a multiscale fusion module to refine feature representation and reduce noise.

To validate the superiority of CausalCD in disentangling the noisy pseudo-label's impact, we also employ five current mature and widely used deep models to construct change detection frameworks through pseudo-label-driven self-training, including LinkNet [46], feature pyramid network (FPN) [47], pyramid attention network (PAN) [48], unified

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perceptual parsing network (UPerNet) [49], and multiscale attention network (MANet) [50]. Additionally, we select three supervised change detection algorithms that also use a pseudo-label generation strategy to obtain training samples.

1) *Spatial-temporal attention-based network (STANet) [51]:* A Siamese neural network with spatio-temporal attention mechanisms, which captures the dependencies in time and space at different scales of paired images through a multiscale self-attention mechanism.
2) *Deeply supervised image fusion network (DSIFN) [52]:* A fully convolutional dual-stream architecture that extracts deep features from multitemporal images. Attention modules integrate multilevel features to improve the accuracy of change regions and introduce change mapping loss at intermediate layers to enhance detection performance.
3) *MiniGCNs [53]:* An innovative GCN designed to train large-scale data in mini-batches efficiently, enhancing model performance by combining the strengths of CNN and GCN.

To ensure a fair comparison with the proposed CausalCD, we generate the training samples for all the aforementioned learning-based algorithms by adopting the same pseudo-label generation strategy. Specifically, the process of generating pseudo-labels is as follows: we utilize FCM to categorize the different image of each test dataset into three classes. These three clusters are sorted based on their mean values, and the cluster with the highest mean pixel values is designated as the changed class, while the cluster with the lowest mean pixel values is designated as the unchanged class. All the learning models were implemented using the source code released by the authors, and it was ensured that the networks converged during the training process.

To quantitatively assess the performance of the change detection algorithms, we employ five evaluation metrics [54]: F-score (F), mean intersection over union (mIoU), false detection rate (FR), missed detection rate (MR), and overall accuracy (OA). They are defined as follows:

$$\begin{aligned}
&\mathrm{P}=\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}},\quad \mathrm{R}=\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}},\quad \mathrm{F}=\frac{\left(1+0.5^2\right)\times \mathrm{P}\times \mathrm{R}}{0.5^2\times \mathrm{P}+\mathrm{R}}\\
&\mathrm{mIoU}=\left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}+\frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}\right)\Big/2\\
&\mathrm{FR}=\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}},\quad \mathrm{MR}=\frac{\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}\\
&\mathrm{OA}=\frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}
\end{aligned}\tag{24}$$

where TP, TN, FP, FN, P, and R represent true positive, true negative, false positive, false negative, precision, and recall, respectively. A higher F, mIOU, and OA are better, while lower FR and MR are preferred. It is noted that we manually annotate the change ground truth based on Google Earth.

C. Comparison to State-of-the-Art Models

1) Quantitative Results: Table II provides a quantitative comparison of different SAR image change detection methods. Several observations can be made from the quantitative results.

1) *Results on Conselice scene:* The town of Conselice in Italy experienced a flood in June 2023, leading to extensive changes between the two images, most notably the inundation of numerous agricultural fields by floodwaters. The experimental results indicate that PCANet, CWNN, STANet, DSIFN, FGDANet, and MiniGCNs all exhibit high FR values. The false detections include some influence of speckle noise, demonstrating that these networks are unable to learn a universal, robust change model from pseudo-labels and are less robust to speckle noise. DCVA, INLPG, IRG-McS, MSDC-SAM, and CausalCD have OA values exceeding 90%, but the F-score and mIoU values of DCVA, INLPG, IRG-McS, and MSDC-SAM are lower than CausalCD, indicating that DCVA, INLPG, IRG-McS, and MSDC-SAM cannot detect all the changes, i.e., it has a high false negative rate. Moreover, although DCVA achieves the lowest FR, its rate of false alarms is higher than that of CausalCD.
2) *Results on Rotterdam scene:* The Rotterdam area in the Netherlands mainly covers the port area, which reveals significant changes in the quantity and movement of ship targets. Hence, despite the temporal proximity of the two images, being only two days apart, numerous changes in ship targets are present. The experimental results reveal that LinkNet, FPN, PAN, UperNet, STANet, and MANet all have relatively high FR values, resulting in lower OA values. While CWNN has the lowest MR at 0.36%, its FR is comparatively high at 6.64%. The proposed CausalCD achieves the best results in the four metrics: F-Score, mIOU, and OA.
3) *Results on Panama Canal scene:* The Panama Canal dataset also covers the port area. Still, the two temporal images are nearly three months apart, leading to changes in ship targets and land. The experimental results indicate that the proposed CausalCD has achieved the optimal F-score, mIoU, and OA values and has lower FR and MR values.

To sum up, the three sets of quantitative experimental results demonstrate that CausalCD consistently achieves the highest F-score, mIoU, and OA values and ensures low false and relatively low miss detections. Some other models may have the lowest miss detections but at the cost of high false detections. Therefore, it is evident that CausalCD outperforms the mainstream change detection models above and has superior change detection performance.

2) Qualitative Results: We plot binary change maps from the detection results obtained by all methods and make a visual comparison, as shown in Figs. 8–10. TP, TN, FP, and FN are presented in white, black, red, and blue, respectively. Visually, the proposed CausalCD performs better than the other methods, with fewer red and blue pixels detected. The results of the first and second datasets show that most models have poor robustness to speckle noise, which results in many false detections of noisy pixels. This is due to incorrect labels in the training samples, which prevent the models from learning effective feature representations. However, the results obtained by the CausalCD are smoother and have fewer noise

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TABLE II

CHANGE DETECTION QUANTITATIVE RESULTS OF DIFFERENT MODELS ON THREE SCENES. THE **BOLD** DENOTES OPTIMAL RESULTS AND THE *Underline* DENOTES THE SECOND BEST RESULTS

Metrics(%)	Conselice					Rotterdam					Panama Canal				
	F↑	mIoU↑	FR↓	MR↓	OA↑	F↑	mIoU↑	FR↓	MR↓	OA↑	F↑	mIoU↑	FR↓	MR↓	OA↑
PCANet(2016) [17]	43.03	54.42	13.67	6.70	79.63	35.68	58.94	9.93	0.96	89.11	39.34	59.82	15.26	4.75	79.99
Linknet(2017) [46]	46.03	55.23	7.34	9.35	83.29	19.26	45.92	21.24	1.36	77.40	54.98	64.13	1.80	3.29	94.91
FPN(2017) [47]	46.85	55.68	7.26	9.22	83.52	24.07	50.77	21.28	1.41	77.31	50.36	63.08	2.49	3.18	94.33
PAN(2018) [48]	58.83	65.71	9.24	4.21	86.55	25.83	52.65	12.27	1.93	85.80	44.07	61.20	3.64	3.07	93.29
UPerNet(2018) [49]	54.19	58.09	4.16	9.73	86.11	16.31	41.51	27.71	1.11	71.18	57.66	65.22	1.63	3.20	95.17
DCVA(2019) [40]	58.99	41.83	**0.07**	9.26	90.67	62.79	45.76	5.38	0.51	94.11	24.93	14.24	**0.03**	5.01	94.96
CWNN(2019) [41]	52.18	61.75	14.21	2.93	82.86	48.75	67.44	6.64	0.36	93.00	49.80	67.15	5.91	0.97	93.12
STANet(2020) [51]	32.97	48.48	11.67	10.36	77.97	29.47	55.07	10.42	1.84	87.74	48.49	61.76	2.34	3.44	94.22
MANet(2020) [50]	56.28	58.76	3.59	9.76	86.65	15.73	42.32	25.10	1.67	73.23	55.43	64.04	1.67	3.35	94.98
DSIFN(2020) [52]	28.76	45.67	15.39	10.18	74.43	36.04	56.05	2.01	4.11	93.88	52.07	63.39	15.27	4.75	79.98
MiniGCNs(2021) [53]	43.32	54.61	16.79	4.95	78.26	53.65	64.59	2.07	2.89	95.04	53.33	63.40	8.21	5.01	86.78
INLPG+PCANet(2021) [42]	65.27	69.24	6.45	4.56	88.99	53.16	69.20	4.79	0.92	94.29	65.87	70.13	1.45	2.60	95.95
INLPG+CWNN(2021) [42]	81.46	74.41	1.26	6.15	92.59	48.75	67.44	6.64	0.36	93.00	53.48	70.04	5.81	**0.41**	93.78
IRG-McS(2021) [43]	81.65	75.13	1.38	5.85	92.77	67.55	71.91	1.42	2.20	96.38	60.82	69.77	2.39	2.20	95.41
MSDC-SAM(2022) [45]	68.28	51.83	2.90	6.19	90.91	59.95	42.81	**1.29**	2.58	96.13	31.81	18.91	0.28	4.69	95.03
FGDANet(2023) [44]	42.58	27.05	36.65	**1.74**	61.61	22.04	12.39	36.72	**0.25**	63.03	18.05	9.92	39.53	1.35	59.12
CausalCD	**83.16**	**77.38**	1.46	5.13	**93.41**	**71.73**	**76.23**	1.53	1.58	**96.89**	**72.54**	**70.16**	0.59	3.01	**96.40**

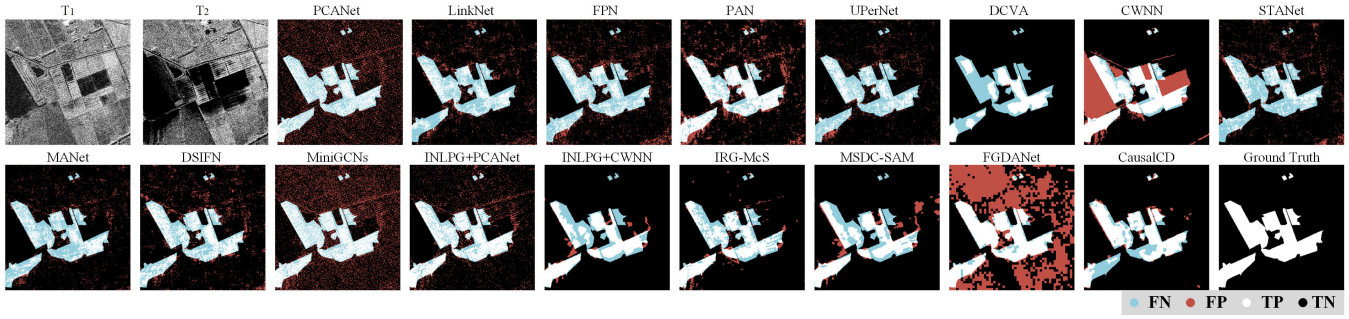


Fig. 8. Comparison of change detection results and ground truth on Conselice scene.

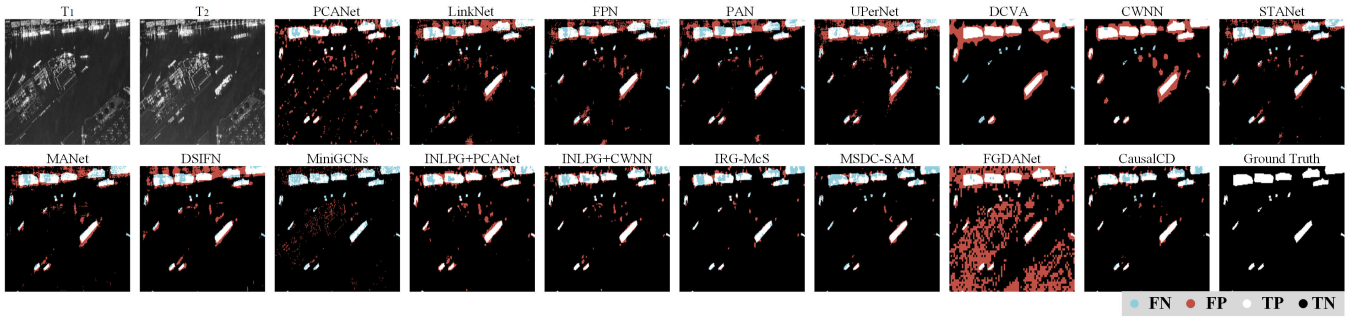


Fig. 9. Comparison of change detection results and ground truth on Rotterdam scene.

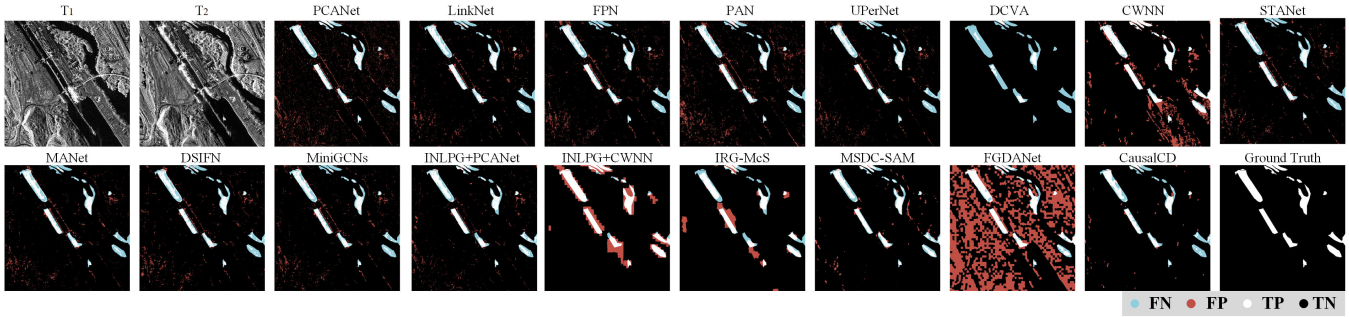


Fig. 10. Comparison of change detection results and ground truth on Panama Canal scene.

points. In the third dataset, most algorithms have significant omissions, while CausalCD shows considerable improvement. Therefore, it can be seen that CausalCD can somewhat eliminate the negative impact of noisy pseudo-labels on the model, resulting in better change detection performance. It can also improve the change detection framework from a causal

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TABLE III
QUANTITATIVE COMPARISON OF GRAPH ENCODER EFFECTIVENESS ANALYSIS

Graph Encoder	Metrics(%)				
	F↑	mIoU↑	FR↓	MR↓	OA↑
Conselice					
RW	67.33	50.75	3.18	6.25	90.57
GK	66.25	49.54	3.43	6.36	90.21
GraphSAGE	**83.16**	**77.38**	**1.46**	**5.13**	**93.41**
Rotterdam					
RW	49.95	56.02	**0.01**	4.56	95.43
GK	64.51	47.61	1.24	**2.28**	96.48
GraphSAGE	**71.73**	**76.23**	1.53	1.58	**96.89**
Panama Canal					
RW	46.58	55.41	0.14	4.90	94.96
GK	49.29	56.94	**0.11**	4.91	94.98
GraphSAGE	**72.54**	**70.16**	0.59	**3.01**	**96.40**

perspective, enhancing the generalizability and rationality of the feature representation model.

D. Analysis of Graph Encoder

In Section IV-D2, we employ a GCL-based framework to extract graph feature representation. Initially, positive and negative augmented graphs are obtained through graph data augmentation, and then a graph encoder is used to extract graph feature representation. In this section, we validate the impact of different graph encoders on change detection performance, adopting RandomWalk (RW) [55], graphlet kernel (GK) [56], and SAmple and aggreGatE (GraphSAGE) [57] as graph encoders. Experiments are conducted on three SAR scenes, and the results are presented in Table III. It can be observed that GraphSAGE achieves the best F-score, mIoU, and OA results on three scenes, showing a significant improvement over RW and GK. This is attributed to GraphSAGE's approach of representing each node by aggregating features from its neighborhood. Hence, even if new nodes not seen during the training process appear in the graph, they can still be represented using their neighboring nodes. Therefore, in the proposed CausalCD framework, GraphSAGE is adopted as the graph encoder.

E. Analysis of Graph Data Augmentation Strategy

In Section IV-D2, we propose a novel causal graph data augmentation strategy from the perspective of causal inference. To verify its effectiveness, we compare it with two other graph data augmentation strategies.

1) Multi-view representation learning on graph (MVGRL) [58] diffuses and samples the original graph to obtain two augmented views.
2) GraphCL [16] uses techniques such as random node dropping, random edge perturbation, random attribute masking, and subgraph to obtain the augmented views of the original graph data.

The aforementioned graph data augmentation strategies are validated on three SAR scenes, with the results illustrated in Fig. 11. It can be observed that the proposed causal graph data augmentation can achieve the best performance on all three SAR image scenes. Specifically, the causal graph data augmentation strategy can generate hard positive and negative instances using graph structural diversity and semantic diversity constraints. On the other hand, the two other data augmentation methods have limitations in instance generation, which produce false positive and negative instances of the original graph, resulting in a decline in change detection model performance.

TABLE IV
FOUR CHANGE DETECTION NETWORKS FOR ABLATION STUDIES TO VERIFY COMPONENT EFFECTIVENESS

Network	CausalGCL	Causal De-Confounding
A	✓	
B		✓
C		
D	✓	✓

F. Analysis of Confounder Set Size

In Section IV-D3, it is necessary to stratify the pseudo-labels to obtain a confounder set, which involves a hyperparameter, specifically the number of stratify, N_C. To investigate the effect of the size of N_C on the performance of CausalCD, we set N_C to 4, 32, 128, and 256, respectively. We then train the CausalCD to obtain the corresponding change detection models and test these models on three scenes. The results are depicted in Fig. 12. The experimental results suggest that the size of the confounder set causes only minor fluctuations in the results. In the first and second scenes, the best results are obtained when the confounder set size is 128, while in the third scene, the optimal results are achieved when the confounder set size is 4.

G. Component Effectiveness

The proposed CausalCD includes two key components: GCL with causal invariant mechanism (CausalGCL) and de-confounded self-training with causal intervention (Causal De-Confounding). To determine the effectiveness of the two components, we consider the four combinations of change detection networks in Table IV.

1) *Network A:* Only the CausalGCL is adopted as the feature representation model, without the Causal De-Confounding to eliminate the negative impact of noisy pseudo-labels on the model.
2) *Network B:* Only the Causal De-Confounding is used to eliminate the negative impact of noisy pseudo-labels, and the feature representation model employs the typical baseline PCANet.
3) *Network C:* The typical baseline PCANet is used as the feature representation model, and the Causal De-Confounding is not employed to eliminate the negative impact of noisy pseudo-labels on the model.
4) *Network D:* The proposed CausalCD in this article uses both the CausalGCL as the feature representation model and Causal De-Confounding to eliminate the negative impact of noisy pseudo-labels.

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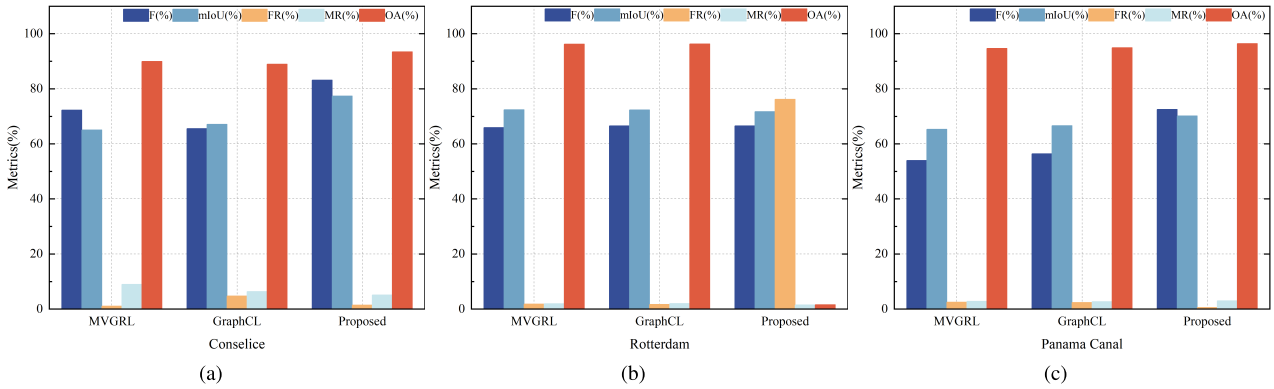


Fig. 11. Validation of the performance of graph data augmentation strategies. (a) Conselice scene. (b) Rotterdam scene. (c) Panama Canal scene.

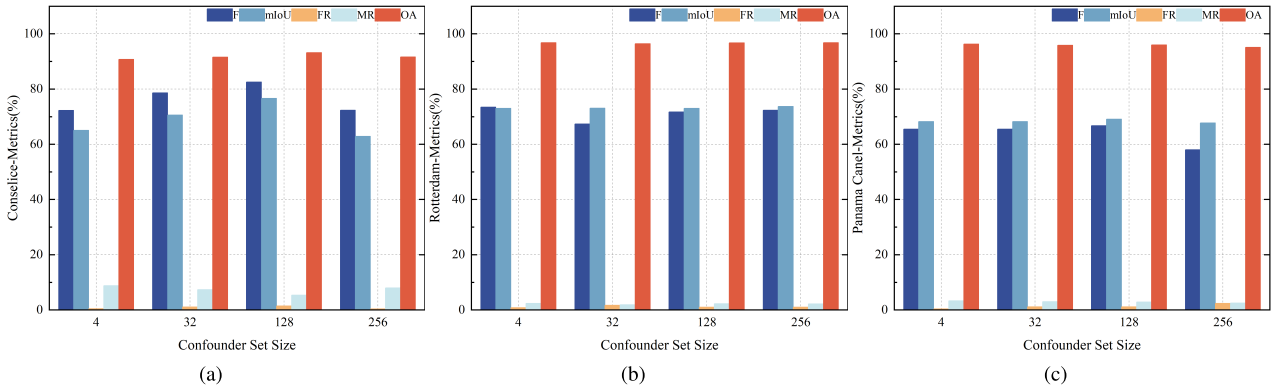


Fig. 12. Validation of the influence of confounder set size N_C. (a) Conselice scene. (b) Rotterdam scene. (c) Panama Canal scene.

We utilize the clustering method described in Section V-B to generate noisy pseudo-label training samples for the four networks and train them to obtain four corresponding change detection models. Afterward, we test on three scenes and calculate the evaluation metrics, as shown in Table V. The experimental results indicate that Networks A and B can obtain effective change detection results, with accuracies surpassing that of Network C. This demonstrates the effectiveness of both components in improving change detection performance. Network D outperforms Networks A and B, indicating that combining CausalGCL with Causal De-Confounding improves change detection by eliminating negative impact of noisy pseudo-labels.

TABLE V

QUANTITATIVE COMPARISON ON THREE SCENES TO VERIFY COMPONENT EFFECTIVENESS

Metrics(%)	Network	Conselice	Rotterdam	Panama Canal
F↑	A	78.16	60.01	63.89
	B	75.39	57.70	58.02
	C	43.03	35.68	39.34
	D	**83.16**	**71.73**	**72.54**
mIoU↑	A	76.01	71.15	66.16
	B	60.50	71.40	67.77
	C	54.42	58.94	59.82
	D	**77.38**	**76.23**	**70.16**
FR↓	A	2.80	2.87	0.91
	B	8.87	3.82	2.40
	C	13.67	9.93	5.38
	D	**1.46**	**1.53**	**0.59**
MR↓	A	4.63	1.56	3.37
	B	**0.94**	1.03	**2.49**
	C	6.70	**0.96**	3.15
	D	5.13	1.58	3.01
OA↑	A	92.57	95.57	95.72
	B	90.19	95.15	95.11
	C	79.63	89.11	91.47
	D	**93.41**	**96.89**	**96.40**

H. Analysis of Parameter Sensitivity

1) Superpixel Size N_S: According to preliminaries in Section IV, prior to the network's self-training, we need to perform superpixel segmentation on bitemporal SAR data to obtain corresponding image patches to construct the graph structures, which serves as the basic input unit of the network. In the superpixel segmentation algorithm SLIC, the parameter N_S determines the level of granularity. A higher N_S value results in finer segmentation and a smaller neighborhood in the constructed graph structure. In order to determine the impact of the size of N_S on the performance of CausalCD, we extract a 50 × 50 image patch on the Panama Canal scene as test data. The N_S values range from 5 to 55, with increments of 5. After training the network under different N_S conditions, we obtain corresponding change detection results as shown in Fig. 13. We can observe that the optimal results for change detection are achieved by setting N_S to 20. This is because when N_S is small, the segmentation granularity is

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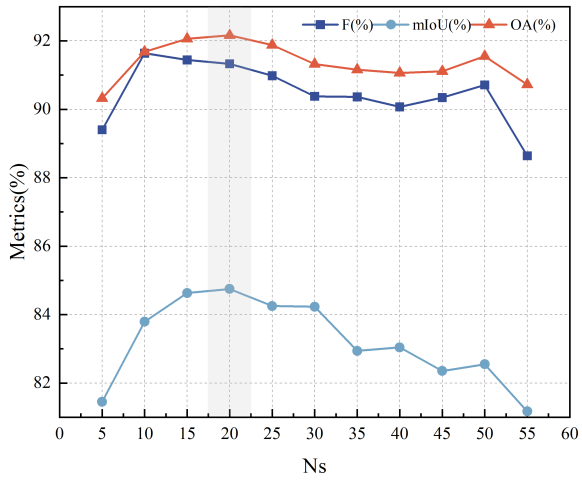


Fig. 13. Quantitative results of change detection at different superpixel sizes.

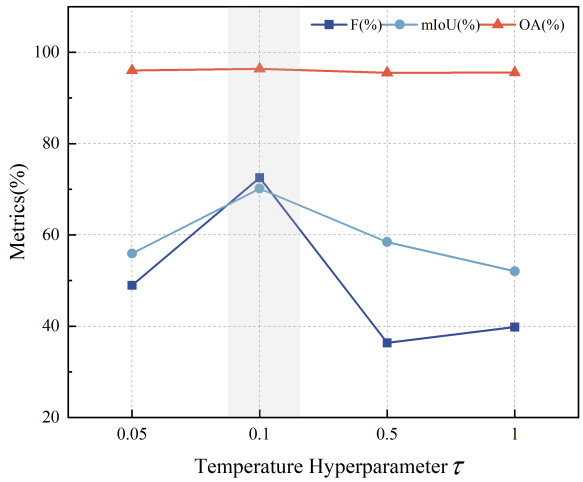


Fig. 14. Quantitative results of change detection at different temperature hyperparameters.

coarse, and the constructed graph structures consider a broader range of neighborhood pixels. However, if the neighborhood range is too large, it may include noise from irrelevant pixels, resulting in decreased performance. On the other hand, when N_S is large, the segmentation granularity is finer, and the constructed graph structures only consider a smaller range of neighborhood pixels. If the neighborhood range is too small, the graph structure fails to contain comprehensive contextual information, making it susceptible to noise, which can negatively affect change detection results. In summary, an appropriate N_S value results in better change detection results. Specifically, when testing with a 50 $\times$ 50 dataset, setting N_S to 20 means that each graph structure takes into account 125 (50 $\times$ 50/20) neighborhood pixels. Therefore, when processing other test datasets, N_S should be adjusted according to the image size to satisfy (height $\times$ width)$/N_S$ is approximately 125.

2) Temperature Hyperparameter τ: In (18), there is an adjustable hyperparameter τ, which has a significant impact on model performance. Therefore, to determine the appropriate value of τ for the proposed CausalCD, we set τ to 0.05, 0.1, 0.5, and 1, respectively. Subsequently, we validate three SAR image scenes, and the results obtained are as illustrated in Fig. 14. It has been observed that when the hyperparameter τ is set to 0.5 and 1, the detection performance significantly deteriorates. This is because, as the temperature hyperparameter increases, the distance between similar samples becomes larger, which weakens the effect of contrastive learning. Therefore, it is crucial to select an appropriate temperature hyperparameter. Based on the experimental results, it is recommended to set τ to 0.1 as it yields the best detection results.

VI. Conclusion

This article represents the first attempt to explore SAR image change detection from a novel perspective: a causal theory-guided self-supervised framework. Specifically, we identify potential issues in the self-supervised SAR image change detection pipeline through theoretical analysis and experimental validation, setting the stage for our research. Next, we formalize the entire process of self-supervised SAR image change detection from a causal relationship standpoint and re-evaluate the aforementioned issues. Building on this analysis, we propose a new change detection framework, CausalCD. We design a novel feature representation module based on causal reasoning that employs a causal invariant mechanism-based GCL approach to derive attributes of optimal augmented representation, thereby improving the generalizability and rationality of the graph feature representation model. We also adopt a causal intervention approach that disentangles the negative impact of confounding biases from noisy pseudo-labels by blocking backdoor paths. Experimental results on three bitemporal SAR scenes showcase the significant performance improvement achieved by the proposed CausalCD. In the field of SAR image change detection, there still exists an unresolved confusion: the same ground targets exhibit different scattering characteristics in SAR images captured from different orbits, leading to pseudo-changes. Current algorithms struggle to eliminate these pseudo-changes, so in our future work, we aim to delve deeper into how to eliminate pseudo-changes caused by orbital differences, starting from the physical scattering mechanisms.

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